Today

- General uncertainty principle
- The simple harmonic oscillator and number state vector space
- Begin topic 4: Three-dimensional systems

3.6 Non-commuting operators and the general uncertainty principle

Next let us consider two Hermitian operators $\hat{A}$ and $\hat{B}$ that may be commuting or not.

Theorem: For any two observables represented by $\hat{A}$ and $\hat{B}$, the uncertainty ΔA in the measurement of $\hat{A}$ and the uncertainty ΔB in the measurement of $\hat{B}$ in any state satisfy

$$(\Delta A)^2(\Delta B)^2 \geq \left(\frac{1}{2i}\langle[\hat{A},\hat{B}]\rangle\right)^2. \tag{3.6.1}$$

Proof: Define $\Psi = (\hat{A} - \langle\hat{A}\rangle)|\psi\rangle$ and $\Phi = (\hat{B} - \langle\hat{B}\rangle)|\psi\rangle$. We then have

$$\begin{aligned}
(\Delta A)^2(\Delta B)^2 &= \langle\Psi|\Psi\rangle\langle\Phi|\Phi\rangle \\
&\geq |\langle\Psi|\Phi\rangle|^2, \quad \text{using the Schwartz inequality} \\
&= \left(\frac{1}{2}(\langle\Psi|\Phi\rangle + \langle\Phi|\Psi\rangle)\right)^2 + \left(\frac{1}{2i}(\langle\Psi|\Phi\rangle - \langle\Phi|\Psi\rangle)\right)^2 \\
&\geq \left(\frac{1}{2i}(\langle\Psi|\Phi\rangle - \langle\Phi|\Psi\rangle)\right)^2 \\
&= \left(\frac{1}{2i}(\langle\psi|(\hat{A} - \langle\hat{A}\rangle)(\hat{B} - \langle\hat{B}\rangle)|\psi\rangle - \langle\psi|(\hat{B} - \langle\hat{B}\rangle)(\hat{A} - \langle\hat{A}\rangle)\psi\rangle)\right)^2 \\
&= \left(\frac{1}{2i}(\langle[\hat{A},\ \hat{B}]\rangle)\right)^2
\end{aligned} \tag{3.6.2}$$

In particular, for $\hat{A} = \hat{x}$ and $\hat{B} = \hat{p}$, we have $[\hat{A},\ \hat{B}] = i\hbar$ and $\Delta x \Delta p \geq \frac{\hbar}{2}$.

Therefore, if the operators representing physical observables do not commute, they cannot be simultaneously measured to arbitrary precession; when one is more precisely measured the other is less precisely measured. Consequently, we cannot construct simultaneous eigenstates (which correspond to definite eigenvalues) of non-commuting observables.

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We have already seen in problem set 4 that the Gaussian wave packet and the ground state of the simple harmonic oscillator are examples of minimum uncertainty states that saturate the bound, $\Delta x \Delta p = \frac{\hbar}{2}$.

Exercise 3.6.1: Consider a physical observable which can take only two possible values and represented by operator $\hat{A}$, and a second physical observable which can, again, take only two possible values and represented by operator $\hat{B}$. The eigenstates $|a_1\rangle$ and $|a_2\rangle$ of $\hat{A}$ and $|b_1\rangle$ and $|b_2\rangle$ of $\hat{B}$ are related by $|a_1\rangle = \frac{3}{5}|b_1\rangle + \frac{4}{5}|b_2\rangle$ and $|a_2\rangle = \frac{4}{5}|b_1\rangle - \frac{3}{5}|b_2\rangle$.

1. A measurement of $\hat{A}$ (i.e., the observable represented by operator $\hat{A}$) produces a_1. What is the state of the system immediately after the measurement?
2. Right after the measurement of $\hat{A}$ in part (1), a following measurement of $\hat{B}$ is done. What are the possible outcomes and with what probability?
3. Right after the measurement of $\hat{B}$ in part (2), a following measurement of $\hat{A}$ is done. What is the probability for obtaining a_1?
4. Question (3) was answered without knowing the outcome in the measurement of $\hat{B}$ in part (2). Suppose b_2 was measured in part (2). What would have been the probability for measuring a_1 in part (3)?

3.7 Time evolution of expectation value of an operator and Ehrenfest's theorem

Consider an observable $\hat{O}$. How does its expectation value in state $|\psi(t)\rangle$ evolve in time? Starting with $\langle\hat{O}\rangle = \langle\psi(t)|\hat{O}|\psi(t)\rangle$, and using the Schrödinger equation $i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$ and its adjoint $-i\hbar\frac{\partial}{\partial t}\langle\psi(t)| = \langle\psi(t)|\hat{H}$, we have

$$\begin{aligned}
\frac{d}{dt}\langle\hat{O}\rangle &= \langle\psi(t)|\frac{\partial\hat{O}}{\partial t}|\psi(t)\rangle + \langle\psi(t)|\hat{O}\frac{\partial}{\partial t}|\psi(t)\rangle + \frac{\partial}{\partial t}(\langle\psi(t))|\hat{O}|\psi(t)\rangle \\
&= \langle\psi(t)|\frac{\partial\hat{O}}{\partial t}|\psi(t)\rangle - \frac{i}{\hbar}\langle\psi(t)|\hat{O}\hat{H}|\psi(t)\rangle + \frac{i}{\hbar}\langle\psi(t))|\hat{H}\hat{O}|\psi(t)\rangle\rangle \\
&= \langle\psi(t)|\frac{\partial\hat{O}}{\partial t}|\psi(t)\rangle - \frac{i}{\hbar}\langle\psi(t)|\hat{O}\hat{H}|\psi(t)\rangle + \frac{i}{\hbar}\langle\psi(t))|\hat{H}\hat{O}|\psi(t)\rangle \\
&= \langle\frac{\partial\hat{O}}{\partial t}\rangle + \frac{i}{\hbar}\langle[\hat{H},\ \hat{O}]\rangle. \qquad (3.7.1)
\end{aligned}$$

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Observe that the expectation value of an operator that does not explicitly depend on time and commutes with the Hamiltonian operator is constant in time. The expression given by (3.7.1) is called Ehrenfest's theorem.

> **Question 3.7.1:** Why is that the expectation value of an operator that does not explicitly depend on time and commutes with the Hamiltonian operator constant in time?

In particular, for $\hat{O} = \hat{\vec{p}}$, and a Hamiltonian that is time-independent,

$$\begin{aligned} \frac{d}{dt}\langle\hat{\vec{p}}\rangle &= \frac{i}{\hbar}\langle[\frac{\hat{\vec{p}}^2}{2m} + V(\hat{\vec{r}}), \hat{\vec{p}}]\rangle = \frac{i}{\hbar}\langle[V(\hat{\vec{r}}), -i\hbar\vec{\nabla}]\rangle = \frac{i}{\hbar}\langle i\hbar\vec{\nabla}V(\hat{\vec{r}})\rangle \\ &= -\langle\vec{\nabla}V(\hat{\vec{r}})\rangle. \end{aligned} \tag{3.7.2}$$

Recalling that $\frac{d}{dt}\vec{p} = -\vec{\nabla}V(\vec{r})$ in classical mechanics, we see that the time evolution of the expectation value of momentum follows the classical equation with the expectation value of the negative gradient of the potential. Observe that $\frac{d}{dt}\vec{p} = -\vec{\nabla}V(\vec{r})$ is just Newton's second law, which governs classical mechanics, and, therefore, quantum mechanics contains all of classical mechanics.

3.8 The simple harmonic oscillator

Consider a particle of mass m in the one-dimensional simple harmonic oscillator. The Hamiltonian operator is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \tag{3.8.1}$$

Can we construct the wavefunctions without solving the Schrödinger equation? The answer is yes. What is the fundamental physics that could allow us to construct the wavefunctions without using the Schrödinger equation? It is the commutation relations between position and momentum. Indeed, one can do all of quantum mechanics starting with the commutation relations and without referring to the Schrödinger equation.

Our starting point now is that we are not going to be attempting to write the Hamiltonian in either the position or the momentum representations. We simply think of $\hat{H}$ as an operator that satisfies commutation relations (with position and momentum operators) that follow from the commutation relations of the position and momentum operators.

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3.8.1 Raising and lowering operators

Suppose we trade $\hat{x}$ and $\hat{p}$ for two other operators $\hat{a}$ and $\hat{a}^\dagger$ such that

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right), \qquad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right). \tag{3.8.2}$$

Note that $\hat{a}$ and $\hat{a}^\dagger$ are not Hermitian, since $\hat{a}^\dagger \neq \hat{a}$. The commutator between $\hat{a}$ and $\hat{a}^\dagger$ is $[\hat{a},\, \hat{a}^\dagger] = \frac{m\omega}{2\hbar}\frac{i}{m\omega}\left([\hat{p}, \hat{x}] - [\hat{x}, \hat{p}]\right) = \frac{i}{2\hbar}\left(-i\hbar - i\hbar\right]) = 1$. Therefore,

$$[\hat{a},\, \hat{a}^\dagger] = 1. \tag{3.8.3}$$

Next, we want to rewrite $\hat{H}$ in terms of $\hat{a}$. First, solving (3.8.2) for $\hat{x}$ and $\hat{p}$,

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}^\dagger + \hat{a}\right), \qquad \hat{p} = i\sqrt{\frac{m\omega\hbar}{2}}\left(\hat{a}^\dagger - \hat{a}\right), \tag{3.8.4}$$

which in (3.8.1) gives

$$\begin{aligned}
\hat{H} &= \frac{1}{2m}\left(i\sqrt{\frac{m\omega\hbar}{2}}\left(\hat{a}^\dagger - \hat{a}\right)\right)^2 + \frac{1}{2}m\omega^2\left(\sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}^\dagger + \hat{a}\right)\right)^2 \\
&= -\frac{1}{4}\hbar\omega\left(\hat{a}^{\dagger 2} + \hat{a}^2 - \hat{a}^\dagger\hat{a} - \hat{a}\hat{a}^\dagger\right) + \frac{1}{4}\hbar\omega\left(\hat{a}^{\dagger 2} + \hat{a}^2 + \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\right) \\
&= \frac{1}{2}\hbar\omega\left(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\right) = \frac{1}{2}\hbar\omega\left(2\hat{a}^\dagger\hat{a} + [\hat{a},\, \hat{a}^\dagger]\right) = \left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)\hbar\omega \\
&= \left(\hat{N} + \frac{1}{2}\right)\hbar\omega,
\end{aligned} \tag{3.8.5}$$

where we have defined

$$\hat{N} = \hat{a}^\dagger\hat{a}. \tag{3.8.6}$$

Now $\hat{N}$ is Hermitian and, therefore, it has orthonormal eigenstates with corresponding eigenvalues. If we solve the eigenvalue problem for the operator $\hat{N}$, we will also have solved the eigenvalue problem for the Hamiltonian operator $\hat{H}$. Therefore, we seek the eigenstate $|n\rangle$ and the corresponding eigenvalue n that solves

$$\hat{N}|n\rangle = n|n\rangle. \tag{3.8.7}$$

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Consider the commutator between $\hat{N}$, $\hat{a}$, and $\hat{a}^\dagger$,

$$[\hat{N},\,\hat{a}] = [\hat{a}^\dagger\hat{a},\,\hat{a}] = \hat{a}^\dagger[\hat{a},\,\hat{a}] + [\hat{a}^\dagger,\,\hat{a}]\hat{a} = -\hat{a} \tag{3.8.8}$$

and

$$[\hat{N},\,\hat{a}^\dagger] = [\hat{a}^\dagger\hat{a},\,\hat{a}^\dagger] = \hat{a}^\dagger[\hat{a},\,\hat{a}^\dagger] + [\hat{a}^\dagger,\,\hat{a}^\dagger]\hat{a} = \hat{a}^\dagger. \tag{3.8.9}$$

Question 3.8.1: $\hat{a}|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue (A). n, (B). $n+1$, (C). $n-1$, (D). $-n$, (D). 0

Now, given $\hat{N}|n\rangle = n|n\rangle$ and $[\hat{N},\,\hat{a}] = -\hat{a}$, we have

$$\hat{N}(\hat{a}|n\rangle) = \left([\hat{N},\,\hat{a}] + \hat{a}\hat{N}\right)|n\rangle = \left(-\hat{a} + \hat{a}\hat{N}\right)|n\rangle = (-\hat{a} + \hat{a}n)\,|n\rangle = (n-1)\,(\hat{a}|n\rangle)\,. \tag{3.8.10}$$

Therefore, $\hat{a}|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue $n-1$. Because $\hat{a}$ applied on $|n\rangle$ lowers the eigenvalue of $\hat{N}$ by one unit, we call $\hat{a}$ a lowering operator. Moreover,

$$\hat{N}\left(\hat{a}^\dagger|n\rangle\right) = \left([\hat{N},\,\hat{a}^\dagger] + \hat{a}^\dagger\hat{N}\right)|n\rangle = \left(\hat{a}^\dagger + \hat{a}^\dagger\hat{N}\right)|n\rangle = \left(\hat{a}^\dagger + \hat{a}^\dagger n\right)|n\rangle = (n+1)\left(\hat{a}^\dagger|n\rangle\right). \tag{3.8.11}$$

Therefore, $\hat{a}^\dagger|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue $n+1$, and we call $\hat{a}^\dagger$ a raising operator.

3.8.2 Normalized number state vectors

Because the energy levels of the simple harmonic oscillator in one dimension are not degenerate, the states $|n-1\rangle$ and $\hat{a}|n\rangle$ must be physically the same, and could only differ by a constant factor. Thus, we write

$$|n-1\rangle = c_n\hat{a}|n\rangle, \tag{3.8.12}$$

where c_n is a constant. The constant c_n follows from the normalization of the states. In particular, considering the matrix element of $\hat{N}$,

$$\langle n|\hat{N}|n\rangle = \langle n|\hat{a}^\dagger\hat{a}|n\rangle = |\hat{a}|n\rangle|^2\,. \tag{3.8.13}$$

On the other hand,

$$\langle n|\hat{N}|n\rangle = n\langle n|n\rangle = n \tag{3.8.14}$$

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for normalized $|n\rangle$. Therefore,

$$1 = \langle n-1|n-1\rangle = |c_n|^2\langle n|\hat{N}|n\rangle = |c_n|^2 n, \tag{3.8.15}$$

or, choosing real c_n, we have $c_n = \frac{1}{\sqrt{n}}$, and

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle. \tag{3.8.16}$$

Similarly,

$$|n+1\rangle = d_n\hat{a}^\dagger|n\rangle, \tag{3.8.17}$$

where d_n is a constant. But

$$1 = \langle n+1|n+1\rangle = |d_n|^2\langle n|\hat{a}\hat{a}^\dagger|n\rangle = |d_n|^2\langle n|[\hat{a},\,\hat{a}^\dagger] + \hat{a}^\dagger\hat{a}|n\rangle = |d_n|^2(1+n). \tag{3.8.18}$$

Therefore, choosing real d_n, we have $d_n = \frac{1}{\sqrt{n+1}}$, and

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \tag{3.8.19}$$

Because the eigenvalue n of $\hat{N}$ satisfies

$$n = \langle n|\hat{N}|n\rangle = |\hat{a}|n\rangle|^2 \geq 0, \tag{3.8.20}$$

it is positive definite or zero. Because $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ and we cannot have negative n, we require $\hat{a}|0\rangle = 0$. Therefore, the ground state is $|0\rangle$. The excited states are then constructed by applying the raising operator on $|0\rangle$. In particular,

$$\hat{a}^\dagger|0\rangle = \sqrt{1}|1\rangle \quad \text{or} \quad |1\rangle = \hat{a}^\dagger|0\rangle, \tag{3.8.21}$$

$$\hat{a}^\dagger|1\rangle = \sqrt{2}|2\rangle \quad \text{or} \quad |2\rangle = \frac{1}{\sqrt{2}}\hat{a}^\dagger|1\rangle = \frac{(\hat{a}^\dagger)^2}{\sqrt{2}}|0\rangle, \tag{3.8.22}$$

$$\hat{a}^\dagger|2\rangle = \sqrt{3}|3\rangle \quad \text{or} \quad |3\rangle = \frac{1}{\sqrt{3}}\hat{a}^\dagger|2\rangle = \frac{(\hat{a}^\dagger)^3}{\sqrt{3!}}|0\rangle, \tag{3.8.23}$$

and continuing,

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \quad n = 0, 1, 2, \ldots \tag{3.8.24}$$

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The matrix elements of position and momentum follow from (3.8.4), (3.8.16), and (3.8.19),

$$\langle n'|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\langle n'|\left(\hat{a}^\dagger + \hat{a}\right)|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n+1}\,\delta_{n',n+1} + \sqrt{n}\,\delta_{n',n-1}\right) \tag{3.8.25}$$

and

$$\langle n'|\hat{p}|n\rangle = i\sqrt{\frac{m\omega\hbar}{2}}\langle n'|\left(\hat{a}^\dagger - \hat{a}\right)|n\rangle = i\sqrt{\frac{m\omega\hbar}{2}}\left(\sqrt{n+1}\,\delta_{n',n+1} - \sqrt{n}\,\delta_{n',n-1}\right). \tag{3.8.26}$$

Note that both $\hat{x}$ and $\hat{p}$ have only off-diagonal matrix elements, which is consistent with $\hat{H}$ or $\hat{N}$ having diagonal elements in the basis with state vectors $|n\rangle$, since $\hat{H}$ does not commute with either one of $\hat{x}$ or $\hat{p}$.

Question 3.8.2: Why should the ground state be $|0\rangle$ with eigenvalue $n = 0$ for $\hat{N}$? Why not a nonzero constant?

Question 3.8.3: Consider a set of operators $\hat{A}$, $\hat{R}$, and $\hat{L}$ such that $\hat{A}|a\rangle = a|a\rangle$, $[\hat{A}, \hat{R}] = r\hat{R}$, and $[\hat{A}, \hat{L}] = -l\hat{L}$ with $l, r > 0$. Which of the following statements are valid. (A). $|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue a. (B). $\hat{L}|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue $a - l$. (C). $\hat{R}|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue $a + r$. (D). $\hat{L}$ is a lowering operator. (E). $\hat{R}$ is a raising operator.

3.8.3 Wavefunctions in position representation

First, the eigenvalues follow from $\hat{H}|n\rangle = \left(\hat{N} + \frac{1}{2}\right)\hbar\omega|n\rangle = \left(n + \frac{1}{2}\right)\hbar\omega|n\rangle$. Therefore,

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad n = 0, 1, 2, \ldots \tag{3.8.27}$$

Next, we want to construct the wavefuctions corresponding to these energy levels in position representation. In particular, the stationary wavefunctions of definite energy are

$$\psi_n(x) = \langle x|n\rangle. \tag{3.8.28}$$

First, let us construct the ground state wavefunction. Using $\hat{a}|0\rangle = 0$,

$$0 \;=\; \hat{a}|0\rangle = \langle x|\hat{a}|0\rangle = \sqrt{\frac{m\omega}{2\hbar}}\langle x|\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)|0\rangle$$

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$$= \sqrt{\frac{m\omega}{2\hbar}}\left(x\langle x|0\rangle + \frac{i}{m\omega}(-i\hbar\frac{\partial}{\partial x})\langle x|0\rangle\right). \tag{3.8.29}$$

Thus, we have a differential equation for $\langle x|0\rangle$,

$$\frac{\hbar}{m\omega}\frac{\partial}{\partial x}\langle x|0\rangle + x\langle x|0\rangle = 0. \tag{3.8.30}$$

The solution is

$$\langle x|0\rangle = \mathcal{A}e^{-\frac{m\omega}{2\hbar}x^2}. \tag{3.8.31}$$

Normalizing,

$$\langle x|0\rangle = \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}. \tag{3.8.32}$$

The wavefunctions for the excited states are then obtained by applying the raising operator. First let us determine the matrix element of $\langle x'|\hat{a}^\dagger|x''\rangle$,

$$\begin{aligned}
\langle x'|\hat{a}^\dagger|x''\rangle &= \langle x'|\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)|x''\rangle = \sqrt{\frac{m\omega}{2\hbar}}\left(\langle x'|\hat{x}|x''\rangle - \frac{i}{m\omega}\langle x'|\hat{p}|x''\rangle\right) \\
&= \sqrt{\frac{m\omega}{2\hbar}}\left(x''\delta(x'-x'') - \frac{i}{m\omega}\left(-i\hbar\frac{\partial\delta(x'-x'')}{\partial x'}\right)\right) \\
&= \frac{1}{\sqrt{2}\sigma}\left(x''\delta(x'-x'') - \sigma^2\frac{\partial\delta(x'-x'')}{\partial x'}\right) \\
&= \frac{1}{\sqrt{2}\sigma}\left(x''\delta(x'-x'') + \sigma^2\frac{\partial\delta(x'-x'')}{\partial x''}\right) \\
&= \delta(x'-x'')\frac{1}{\sqrt{2}\sigma}\left(x'' - \sigma^2\frac{\partial}{\partial x''}\right),
\end{aligned} \tag{3.8.33}$$

where we have defined

$$\sigma \equiv \sqrt{\frac{\hbar}{m\omega}} \tag{3.8.34}$$

for simplicity of notation. Inserting the above expression for the matrix elements and integrating,

$$\begin{aligned}
\langle x|n\rangle &= \langle x|\frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle = \frac{1}{\sqrt{n!}}\int dx_1 dx_2 ... dx_{n-1}\langle x|\hat{a}^\dagger|x_1\rangle\langle x_1|\hat{a}^\dagger|x_2\rangle\langle x_2|\cdots|x_{n-1}\rangle\langle x_{n-1}|\hat{a}^\dagger|x_n\rangle\langle x_n|0\rangle \\
&= \frac{1}{\sqrt{n!}}\left(\frac{1}{\sqrt{2}\sigma}\left(x - \sigma^2\frac{\partial}{\partial x}\right)\right)^n\langle x|0\rangle \\
&= \frac{1}{\sqrt{n!2^n}\sigma^n}\left(x - \sigma^2\frac{\partial}{\partial x}\right)^n\left(\frac{1}{\pi\sigma^2}\right)^{1/4} e^{-\frac{1}{2\sigma^2}x^2}
\end{aligned} \tag{3.8.35}$$

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$$= \frac{1}{\sqrt{\sqrt{\pi} n! 2^n \, \sigma^{n+\frac{1}{2}}}} \left(x - \sigma^2 \frac{\partial}{\partial x} \right)^n e^{-\frac{1}{2\sigma^2} x^2}. \tag{3.8.36}$$

If we define

$$\xi = \frac{x}{\sigma}, \tag{3.8.37}$$

then

$$\langle x|n\rangle = \frac{1}{\sqrt{\sqrt{\pi} n! 2^n \sigma}} \left(\xi - \frac{\partial}{\partial \xi} \right)^n e^{-\frac{1}{2}\xi^2} \tag{3.8.38}$$

This reproduces the wavefunctions obtained after solving the Schrodinger equation. In particular, for the ground and first excited states,

$$\langle x|0\rangle = \frac{1}{\sqrt{\sqrt{\pi}\sigma}} e^{-\frac{1}{2}\xi^2} = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega}{2\hbar} x^2} \tag{3.8.39}$$

and

$$\langle x|1\rangle = \frac{1}{\sqrt{\sqrt{\pi} 2\sigma}} \left(\xi - \frac{\partial}{\partial \xi} \right) e^{-\frac{1}{2}\xi^2} = \frac{1}{\sqrt{\sqrt{\pi} 2\sigma}} 2\xi e^{-\frac{1}{2}\xi^2} = \sqrt{2} \left(\frac{m^3\omega^3}{\pi\hbar^3} \right)^{1/4} x e^{-\frac{m\omega}{2\hbar} x^2}. \tag{3.8.40}$$

3.8.4 Classical simple harmonic oscillator

The probability density for finding the particle at point x in a stationary states $\langle x|n\rangle$ does not evolve in time. In other words, it does not oscillate. Can we construct a quantum state that exhibits a classical-like oscillation?

First let us briefly overview the classical equations of motion of the simple harmonic oscillator in terms of variables that would resemble the quantum oscillator. The Hamiltonian (or total energy) for a classical simple harmonic is

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2, \tag{3.8.41}$$

which is the same as (3.8.1) before the position and momentum are promoted to operators. The

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time evolution of x and p are given by Hamilton's equations,

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m} \tag{3.8.42}$$

and

$$\dot{p} = -\frac{\partial H}{\partial x} = -m\omega^2 x. \tag{3.8.43}$$

Let us trade x and p for a new complex variable α defined by

$$\sqrt{\hbar\omega}\,\alpha = \sqrt{\frac{m\omega^2}{2}}x + \frac{i}{\sqrt{2m}}p. \tag{3.8.44}$$

With this, we rewrite the Hamiltonian,

$$H = \hbar\omega|\alpha|^2. \tag{3.8.45}$$

Moreover,

$$x = \sqrt{\frac{2\hbar}{m\omega}}\,\alpha_R, \qquad p = \sqrt{2m\hbar\omega}\,\alpha_I \tag{3.8.46}$$

where α_R and α_I are the real and imaginary parts of α.

The time evolution of α is then

$$\begin{aligned}
\sqrt{\hbar\omega}\,\dot{\alpha} &= \sqrt{\frac{m\omega^2}{2}}\dot{x} + \frac{i}{2m}\dot{p} = \sqrt{\frac{m\omega^2}{2}}\frac{p}{m} + \frac{i}{\sqrt{2m}}(-m\omega^2 x) \\
&= \sqrt{\frac{m\omega^2}{2}}\frac{1}{m}\sqrt{2m\hbar\omega}\,\alpha_I - \frac{i}{\sqrt{2m}}m\omega^2\sqrt{\frac{2\hbar}{m\omega}}\,\alpha_R \\
&= -i\sqrt{\hbar\omega^3}\,(\alpha_R + i\alpha_I) = -i\sqrt{\hbar\omega^3}\,\alpha.
\end{aligned} \tag{3.8.47}$$

or

$$\dot{\alpha} = -i\omega\alpha. \tag{3.8.48}$$

The solution is

$$\alpha = \alpha_0\, e^{-i\omega t}. \tag{3.8.49}$$

As a specific example for a real α_0, then $\alpha_R = \alpha_0\cos\omega t$ and $\alpha_I = \alpha_0\sin\omega t$. In this case, $x = \sqrt{\frac{2\hbar}{m\omega}}\,\alpha_R = \sqrt{\frac{2\hbar}{m\omega}}\,\alpha_0\cos\omega t$ and $p = \sqrt{2m\hbar\omega}\,\alpha_I = \sqrt{2m\hbar\omega}\,\alpha_0\sin\omega t$. Therefore, the real and imaginary parts of α are related to the position and momentum of the oscillator. In particular,

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expressing the Hamiltonian of the classical oscillator in terms of α,

$$H = |\alpha|^2 \hbar\omega. \tag{3.8.50}$$

3.8.5 The quantum simple harmonic oscillator and coherent state

Question 3.8.4: What is the frequency of oscillation of the probability density for a simple harmonic oscillator in state $|n\rangle$? (A). ω, (B). $\frac{E_n}{\hbar}$, (C). $\frac{E_n - E_{n-1}}{\hbar}$, (D). 0.

Question 3.8.5: What could we do to construct a state whose time evolution could look like a classical oscillator?

Returning back to the quantum oscillator, and noting the expression $H = |\alpha|^2\hbar\omega$ for the classical oscillator, we observe the expectation value of $\hat{H}$,

$$\langle \hat{H} \rangle = \left(\langle \hat{a}^\dagger \hat{a} \rangle + \frac{1}{2} \right) \hbar\omega \tag{3.8.51}$$

has a similar form as the classical H if we identify $\langle \hat{a}^\dagger \hat{a} \rangle$ with $|\alpha|^2$ and redefine the classical potential energy such that $H \to H = \left(|\alpha|^2 + \frac{1}{2} \right) \hbar\omega$. Thus, we seek a quantum state $|\alpha\rangle$ which gives $\langle \alpha | \hat{a}^\dagger \hat{a} | \alpha \rangle = |\alpha|^2$. This is possible if we could construct a state $|\alpha\rangle$ such that

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle. \tag{3.8.52}$$

The state $|\alpha\rangle$ that satisfies this equation is called coherent state (or Glauber state, after the recently late Roy Glauber of Harvard). It is a quantum state whose dynamics resembles that of a classical oscillator. We will investigate it in problem set 6.

Question 3.8.6: What is the frequency of oscillation of the probability density of a coherent state of energy $\frac{1}{2}\hbar\omega$? What about for a coherent state of energy $E > \frac{1}{2}\hbar\omega$?

Exercise 3.8.1: Using the raising and lowering operators, answer the following questions for a particle of mass m in the one-dimensional harmonic oscillator with potential $V(x) = \frac{1}{2} m\omega^2 x^2$.

1. The uncertainty in position, Δx, in stationary state $|n\rangle$.
2. The uncertainty in momentum, Δp, in stationary state $|n\rangle$.
3. Check whether the product $\Delta x \, \Delta p$ obeys the uncertainty principle.

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4. Rewrite the Hamiltonian $\hat{H}$ in terms of $\hat{\alpha}_R$ and $\hat{\alpha}_I$ defined by $\hat{x} = \sqrt{\frac{2\hbar}{m\omega}}\,\hat{\alpha}_R$ and $\hat{p} = \sqrt{2m\hbar\omega}\,\hat{\alpha}_I$. Rewrite the uncertainty relation in stationary state $|n\rangle$ obtained in part (3) for $\Delta\alpha_R \Delta\alpha_I$.

5. When the oscillator is not in a stationary state, or an eigenstate of the Hamiltonian operator, the uncertainty would take other values than those for individual stationary states obtained above. Show the boundary that separates the region that satisfies the uncertainty principle and the region that does not on a plot of $\Delta\alpha_I$ versus $\Delta\alpha_R$. Indicate the points $(\Delta\alpha_R, \Delta\alpha_I)$ on the plane for the first four stationary states corresponding to $n = 0, 1, 2$, and 3.

Exercise 3.8.2: A coherent state (or Glauber state) $|\alpha\rangle$ of the simple harmonic oscillator is an eigenstate of the lowering operator, $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$.

1. Determine the coherent state as a superposition of the stationary states $|n\rangle$ and normalize it.

2. What is the probability that a system in a coherent state would be found in stationary state $|n\rangle$? What kind of probability distribution function of n is it?

3. Determine the uncertainties Δx, $\Delta\alpha_R$, Δp, and $\Delta\alpha_R$, using the raising and lowering operators, in the coherent state.

4. Show the point for uncertainties $(\Delta\alpha_R, \Delta\alpha_I)$ for the coherent state on the plot you made in part (5). Comparing $\Delta\alpha_R$, $\Delta\alpha_I$, and their product, what is special about the coherent state?

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4 Three-dimensional systems

Now, we begin investigating the Schrödinger equation for a particle in three dimensions with time-independent potential $V(\vec{r})$.

4.1 Three-dimensional infinite square well

Consider a particle of mass m in potential $V(x, y, z) = 0$ for $0 \leq x \leq l_x$, $0 \leq y \leq l_y$, $0 \leq z \leq l_z$, and $V(x, y, z) = \infty$ outside. Because the potential is cubic, the Schrödinger equation in cartesian coordinate is suitable. Therefore, the time-independent Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi(x,y,z) = E\psi(x,y,z) \tag{4.1.1}$$

for $0 \leq x \leq l_x$, $0 \leq y \leq l_y$, and $0 \leq z \leq l_z$, while $\psi(x, y, z) = 0$ outside. Because there is no potential that mixes the different coordinates, we can use the method of separation of variables, and start with writing,

$$\psi(x, y, t) = \psi_1(x)\psi_2(y)\psi_3(z). \tag{4.1.2}$$

The Schrödinger equation then becomes

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi_1(x) = E_1\psi_1(x), \quad -\frac{\hbar^2}{2m}\frac{d^2}{dy^2}\psi_2(y) = E_2\psi_2(y), \quad -\frac{\hbar^2}{2m}\frac{d^2}{dz^2}\psi_3(z) = E_3\psi_3(z), \tag{4.1.3}$$

where

$$E = E_1 + E_2 + E_3. \tag{4.1.4}$$

We know the solutions from the one-dimensional infinite square well,

$$\psi_{n_x n_y n_z}(x, y, t) = \sqrt{\frac{8}{l_x l_y l_z}}\sin(\frac{n_x\pi}{l_x}x)\sin(\frac{n_y\pi}{l_y}z)\sin(\frac{n_z\pi}{l_z}z) \tag{4.1.5}$$

and

$$E_{n_x n_y n_z} = \frac{\hbar^2\pi^2}{2m}\left(\frac{n_x^2}{l_x^2}+\frac{n_y^2}{l_y^2}+\frac{n_z^2}{l_z^2}\right) \tag{4.1.6}$$

with $n_x, n_y, n_z = 1, 2, ...$ Note that the wave vector is

$$\vec{k} = (k_x, k_y, k_z) = (\frac{n_x\pi}{l_x}, \frac{n_y\pi}{l_y}, \frac{n_z\pi}{l_z}). \tag{4.1.7}$$

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Exercise 4.1.1: When we talked about Planck's black-body radiation formula, we mentioned that the number density (per unit volume per unit frequency) of states in a cavity was $\frac{\mathcal{D}(\omega)}{V} = \frac{\omega^2}{\pi^2 c^3}$. Show that considering the standing modes of the three-dimensional infinite square well in a large cubic box.

Solution: Considering a large box, we can view its volume in $\vec{k}$–space as spherical, and the volume of spherical shell of infinitesimal thickness dk at radius k is $\frac{1}{8} 4\pi k^2 dk = \frac{1}{2}\pi k^2 dk$, where the factor of $\frac{1}{8}$ because all components of $\vec{k}$ are positive. The volume of a single cell is $\vec{k}$–space is $\frac{\pi}{l_x}\frac{\pi}{l_y}\frac{\pi}{l_z} = \frac{\pi^2}{V}$, where $V = l_x l_y l_z$ is the volume of the box. The number of states within this volume is $dN = 2\frac{\frac{1}{2}\pi k^2 dk}{\frac{\pi^3}{V}} = \frac{V}{\pi^2} k^2 dk$, where the factor of 2 is because a standing plane wave has two independent directions (for describing the plane on which the electric field lives). Considering radiation, $k = \frac{\omega}{c}$. Thus, $dN = 2\frac{V}{\pi^2}\frac{\omega^2}{c^2}\frac{d\omega}{c}$ and $\frac{\mathcal{D}(\omega)}{V} = \frac{dN}{d\omega V} = \frac{\omega^2}{\pi^2 c^3}$.

Exercise 4.1.2: The Hamiltonian operator for a particle of mass m constrained to undergo circular motion on a frictionless track is given by $\hat{H} = -\frac{\hbar^2}{2mR^2}\frac{\partial^2}{\partial\phi^2}$, where R is the radius of the circle and ϕ is the angular coordinate. What boundary condition should the wavefunction $\psi(\phi)$ satisfy? Using the Schrödinger equation and the boundary condition, determine the wavefunctions and energies of the stationary states. Determine the degeneracy of each energy level.

Exercise 4.1.3: Consider the three-dimensional harmonic oscillator with potential $V(x, y, z) = \frac{1}{2}m\omega^2(x^2 + y^2 + z^2)$. Answer the following questions using cartesian coordinates, separation of variables, and your knowledge of the energies and wavefunctions of the one-dimensional simple harmonic oscillator. Write down the energy levels in terms of three quantum numbers n_1, n_2, and n_3. Write the same energy levels in terms of one quantum number n. Show that the nth energy level is $\frac{1}{2}(n+1)(n+2)$–fold degenerate.

4.2 The Schrödinger equation in spherical coordinates

The Schrödinger equation in three dimensions in position representation in spherical coordinates is

$$i\hbar\frac{\partial\psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}, t) + V(\vec{r})\psi(\vec{r}, t), \tag{4.2.1}$$

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where $\vec{r} = (r, \theta, \phi)$, $\psi(\vec{r}, t) = \psi(r, \theta, \phi, t)$ and

$$\vec{\nabla}^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} \tag{4.2.2}$$

is the Laplacian operator.

For a time-independent and central potential, $V(\vec{r}) = V(r)$. In other words, the potential depends only of the radial coordinate r but not on θ and ϕ. The time-independent Schrödinger equation then becomes

$$-\frac{\hbar^2}{2mr^2}\left(\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi(r,\theta,\phi)}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi(r,\theta,\phi)}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi(r,\theta,\phi)}{\partial\phi^2}\right) + V(r)\psi(r,\theta,\phi) = E\psi(r,\theta,\phi). \tag{4.2.3}$$

Because the potential does not mix the different coordinates, we can make separation of variables and write

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi). \tag{4.2.4}$$

The Schrödinger equation then reduces to

$$\frac{1}{R(r)}\left[\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) - \frac{2mr^2}{\hbar^2}(V(r) - E)\right]R(r) = -\frac{1}{Y(\theta,\phi)}\left[\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dY(\theta,\phi)}{d\theta}\right) + \frac{1}{\sin^2\theta}\frac{d^2Y(\theta,\phi)}{d\phi^2}\right]. \tag{4.2.5}$$

Each side must be a constant and equal. Thus, denoting the constant by $l(l+1)$ for convenience in notation later,

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y(\theta,\phi)}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y(\theta,\phi)}{\partial\phi^2} = -l(l+1)Y(\theta,\phi). \tag{4.2.6}$$

and

$$\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) - \frac{2mr^2}{\hbar^2}(V(r) - E)R(r) = l(l+1)R(r). \tag{4.2.7}$$

4.3 Orbital angular momentum

The orbital angular momentum in classical mechanics is defined by $\vec{L} = \vec{r}\times\vec{p}$. In quantum mechanics, this is promoted to an operator, and in position representation,

$$\hat{\vec{L}} = \hat{\vec{r}}\times\hat{\vec{p}}. \tag{4.3.1}$$

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In cartesian coordinates, the components of the angular momentum operator are

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \quad \hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \quad \hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x, \tag{4.3.2}$$

where we need to distinguish the hat over an operator versus over a unit vector from context. Because the components of the position and the momentum operators do not commute, the components of the angular momentum operator do not commute either. In particular,

$$\begin{aligned} [\hat{L}_x, \hat{L}_y] &= [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z] = \hat{y}[\hat{p}_z, \hat{z}]\hat{p}_x] + \hat{x}[\hat{z}, \hat{p}_z]\hat{p}_y \\ &= -i\hbar\hat{y}\hat{p}_x + i\hbar\hat{x}\hat{p}_y = i\hbar\hat{L}_z \end{aligned} \tag{4.3.3}$$

Similarly, and collecting,

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z, \quad [\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_z, \quad [\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y, \tag{4.3.4}$$

or

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k \tag{4.3.5}$$

with $i = 1, 2, 3$ representing the x, y, and z components respectively, and ϵ_{ijk} is the epsilon tensor with components defined by

$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1, \tag{4.3.6}$$

it becomes -1 for odd permutation of indices, such as $\epsilon_{213} = -1$, and vanishes when any index is repeated, such as $\epsilon_{112} = 0$.

Furthermore, the square of the total angular momentum operator is

$$\hat{\vec{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \tag{4.3.7}$$

and it commutes with each component of the total angular momentum. For instance,

$$\begin{aligned} [\hat{\vec{L}}^2, \hat{L}_z] &= [\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \hat{L}_z] = \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x + \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y \\ &= -i\hbar\hat{L}_x\hat{L}_y - i\hbar\hat{L}_y\hat{L}_x + i\hbar\hat{L}_y\hat{L}_x, + i\hbar\hat{L}_x\hat{L}_y = 0. \end{aligned} \tag{4.3.8}$$

Thus,

$$[\hat{\vec{L}}^2, \hat{L}_i] = 0. \tag{4.3.9}$$

Question 4.3.1: Out of the four operators, $\hat{\vec{L}}$, $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$, how many could

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we simultaneously diagonalize? (A). 0, (B). 1, (C). 2, (D). 2, (E). 4.

Question 4.3.2: Out of the four operators, $\hat{\vec{L}}^2$, $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$, how many could we simultaneously diagonalize? (A). 0, (B). 1, (C). 2, (D). 2, (E). 4.

In position representation,

$$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}} = -i\hbar \vec{r} \times \vec{\nabla} \tag{4.3.10}$$

In spherical coordinates,

$$\begin{aligned} \hat{\vec{L}} &= -i\hbar r\hat{r} \times \left(\hat{r}\frac{\partial}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial}{\partial \phi} \right) \\ &= -i\hbar \left(\hat{\phi}\frac{\partial}{\partial \theta} - \hat{\theta}\frac{1}{\sin\theta}\frac{\partial}{\partial \phi} \right). \end{aligned} \tag{4.3.11}$$

The components along cartesian unit vectors can be obtained using the unit vector relations,

$$\hat{r} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z} \tag{4.3.12}$$

$$\hat{\theta} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z} \tag{4.3.13}$$

$$\hat{\phi} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}. \tag{4.3.14}$$

With these,

$$\begin{aligned} \hat{\vec{L}} &= -i\hbar \left((-\sin\phi\,\hat{x} + \cos\phi\,\hat{y})\frac{\partial}{\partial \theta} - (\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z})\frac{1}{\sin\theta}\frac{\partial}{\partial \phi} \right) \\ &= L_x\hat{x} + L_y\hat{y} + L_z\hat{z}, \end{aligned} \tag{4.3.15}$$

where

$$\hat{L}_x = i\hbar \left(\sin\phi\,\frac{\partial}{\partial \theta} + \cot\theta\cos\phi\,\frac{\partial}{\partial \phi} \right) \tag{4.3.16}$$

$$\hat{L}_y = i\hbar \left(-\cos\phi\,\frac{\partial}{\partial \theta} + \cot\theta\sin\phi\,\frac{\partial}{\partial \phi} \right) \tag{4.3.17}$$

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial \phi} \tag{4.3.18}$$

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Squaring, $\hat{\vec{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$,

$$\hat{\vec{L}}^2 = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right] \tag{4.3.19}$$

With this, the angular part of the Schrödinger equation given by (4.2.6) can be rewritten as

$$\hat{\vec{L}}^2 Y(\theta,\phi) = l(l+1)\hbar^2 Y(\theta,\phi). \tag{4.3.20}$$

We also rearrange and rewrite the radial part given by (4.2.7) as

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dR(r)}{dr} \right) + \left(V(r) + \frac{l(l+1)\hbar^2}{2mr^2} \right) R(r) = ER(r). \tag{4.3.21}$$

Or, rewriting again,

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dR(r)}{dr} \right) - V_{\text{eff}}(r)R(r) = ER(r), \tag{4.3.22}$$

which is a Schrödinger like equation for the radial part of the wavefunction with effective potential,

$$V_{\text{eff}}(r) = V(r) + \frac{l(l+1)\hbar^2}{2mr^2}. \tag{4.3.23}$$

Note that the second term is the potential corresponding to centrifugal force in classical mechanics due to the orbital motion of the particle.

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